

Machine Learning

1. Define Machine Learning in your own words.
Explain how it is different from traditional rule-based programming with a real-world example.

2. What are **Independent Variables** and **Dependent Variables** in Machine Learning?
Using the Ball classification case study, clearly identify:

- Independent variables
- Dependent variable

3. Explain the difference between Features and Labels.
Are features always numeric? Justify your answer.

4. What is Supervised Machine Learning?
Why is labeled data mandatory in supervised learning?

5. Explain why we cannot train and test a model using the same dataset.
What problem may occur if we do so?

6. What is the purpose of splitting a dataset into training and testing sets?
1. Define machine Learning in your own words.
Explain how it is different from traditional rule-based programming with a real-world example.

2. What are **Independent Variables** and **Dependent Variables** in Machine Learning?
Using the Ball classification case study, clearly identify:

- Independent variables
- Dependent variable

3. Explain the difference between Features and Labels.
Are features always numeric? Justify your answer.

4. What is Supervised Machine Learning?
Why is labeled data mandatory in supervised learning?

5. Explain why we cannot train and test a model using the same dataset.
What problem may occur if we do so?

6. What is the purpose of splitting a dataset into training and testing sets?
1. Define machine Learning in your own words.
Explain how it is different from traditional rule-based programming with a real-world example.

2. What are **Independent Variables** and **Dependent Variables** in Machine Learning?
Using the Ball classification case study, clearly identify:

- Independent variables
- Dependent variable

3. Explain the difference between Features and Labels.
Are features always numeric? Justify your answer.

4. What is Supervised Machine Learning?
Why is labeled data mandatory in supervised learning?

5. Explain why we cannot train and test a model using the same dataset.
What problem may occur if we do so?

6. What is the purpose of splitting a dataset into training and testing sets?

What is the typical ratio used in industry?